EECSA: Enhancing Textual Sentiment Analysis through Emotion-Enriched Contextual Integration and Self-Adaptive Prompting Optimization*

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***Abstract*—Textual sentiment analysis remains a crucial task in natural language processing (NLP) that focuses on identifying sentiments towards specific entities within a text. Recently, large language models (LLMs) have demonstrated remarkable capabilities in understanding semantics and logical reasoning. However, LLMs often face challenges such as hallucinations, which can lead to inaccurate sentiment determination for target entities. Additionally, current methods frequently fail to fully exploit the extensive prior knowledge embedded within LLMs, resulting in suboptimal sentiment classification. To address these challenges, we propose the Emotion-Enriched Contextual Sentiment Analysis (EECSA) framework. EECSA integrates unique emotion-enriched constraints, backgrounds, and analogical reasoning to mitigate LLM hallucinations and employs self-adaptive bootstrap instructions optimization to further enhance LLM predictions. Specifically, constraints are constructed through a linguistic consistency mechanism and multiple inquiry rounds; background descriptions are incorporated as prompts for LLM inference; and analogies are generated by LLMs to enhance textual sentiment analysis. EECSA further refines initial LLM instructions through adaptive iterative optimization using a random search bootstrap algorithm, maximizing the effectiveness of LLM prompting. Extensive experiments of zero-shot and few-shot learning with `GPT-3.5-turbo` across six public datasets validate the superiority of our EECSA framework, demonstrating performance that in some cases surpasses `GPT-4.0`, the most advanced large language model in the world. Code is available at: https://github.com/albert-jin/EECSA.**

***Index Terms*—textual sentiment analysis, large language model, LLM prompting engineering, self-adaptive bootstrap instructions optimization**

I. Introduction

Textual sentiment analysis (TSA) [1], [2] is crucial in natural language processing (NLP) for identifying sentiment polarities towards specific target entities within a text, thereby enhancing decision-making in marketing, product development, and customer service. For instance, a professional TSA system can provide detailed insights into opinions on specific features such as food quality or service in a restaurant review, helping food vendors optimize their offerings based on user preferences.

Traditional deep learning (DL)-based TSA methods, which primarily rely on pretrained word embeddings and fine-tuning small language models (SLMs) like BERT [3], RoBERTa [4], and T5 [5], face several challenges. They require large amounts of domain-specific training data, which is difficult and costly to obtain. Despite reducing the need for manual feature engineering, these models still demand extensive preprocessing and feature engineering to enhance performance. Additionally, they struggle with domain transferability and multilingual environments, and their relatively small parameter size limits their effectiveness in capturing nuanced sentiments.

The adoption of large language models (LLMs) has emerged as a powerful technique to enhance TSA tasks. LLMs like OpenAI's ChatGPTs [6], [7] and MetaAI's Llama series [8] have shown potential to overcome these limitations. Their impressive language abilities have sparked curiosity about their performance across various domains. Prompting techniques, which equip LLMs with rich vocabulary and advanced response mechanisms, enable better understanding of human intentions and more effective communication. Techniques like Chain-of-Thought (CoT) [9], [10], In-Context Learning (ICL) [11], [12], and Reasoning and Acting (ReAct) [13] have significantly improved LLMs' handling of complex linguistic reasoning tasks. CoT prompts LLMs to engage in step-by-step reasoning, ICL enhances model understanding through examples, and ReAct combines reasoning with actionable steps.

Although representative prompting techniques are effective in enhancing LLM response potential, these methods still have limitations, particularly in specific downstream applications. Techniques such as Chain-of-Thought (CoT) prompting [9],

[10], although beneficial for step-by-step reasoning, suffer from being static and opaque. This lack of control in the reasoning process can lead to issues like factual hallucinations and error propagation due to reliance on internally generated ideas. Similarly, In-Context Learning (ICL) [11], [12], which involves presenting annotated input-label pairs from training data, is highly dependent on the quality of these annotations. Poorly annotated or randomized labels can significantly degrade the performance of LLM predictions, underscoring the need for more robust and context-aware prompting strategies. Additionally, a critical challenge is the insufficient extraction of prior knowledge embedded within LLMs [14], [15]. This underutilization of LLMs' extensive prior knowledge results in sub-optimal classification outcomes, limiting the effectiveness of textual sentiment analysis.

To address these challenges, we propose the Emotion-Enriched Contextual Sentiment Analysis (EECSA) framework. EECSA integrates Emotion-Enriched Contexts, which include Emotion-Enriched Constraints, Backgrounds, and Analogies, to mitigate issues such as hallucinations and further enhance LLM predictions.

- *Emotion-Enriched Constraints* are constructed by generating variant queries that hypothesize different polarities for target entities within a sentence. We then evaluate these responses using a consistency alignment mechanism to ensure accuracy and reduce ambiguity.
- For *Emotion-Enriched Backgrounds*, we prompt LLMs to generate brief introductions for target entities in a sentence, providing additional context for sentiment identification [16]. These background descriptions help in achieving a more nuanced and context-aware sentiment analysis.
- Inspired by human analogical reasoning, we introduce *Emotion-Enriched Analogies*, which prompts LLMs to generate relevant analogical examples [17]. These examples are emotionally and contextually aligned with the given problem, guiding the LLM to better solve the task by leveraging similar past experiences.

All these procedures are implemented through two primary steps: 1. *Emotion-Enriched Context Acquisition*, which involves obtaining emotion-enriched constraints, backgrounds, and analogies; 2. *Emotion Context Integration & Sentiment Prediction*, which focuses on integrating the acquired emotion contexts through In-Context Learning to predict the final sentiment polarity.

Our contributions are summarized as follows:

- We systematically examine the challenges of current LLM prompting methods and propose the Emotion-Enriched Contextual Sentiment Analysis (EECSA) framework. By integrating emotional constraints, backgrounds, and analogies, this framework fully leverages prior knowledge within LLMs, addressing hallucinations and improving sentiment analysis performance.
- We design a *self-adaptive bootstrap instructions optimization* strategy for further optimizng EECSA, fine-tuning prompts to generate more accurate sentiment predictions.
- Comprehensive zero-shot and few-shot experiments with GPT-3.5-turbo across six public datasets validate the effectiveness and robustness of EECSA, surpassing vanilla GPT-4.0 predictions.
- To facilitate further research on textual sentiment analysis, we released the code and data of EECSA on GitHub[1]

The remainder of this paper is organized as follows. Section II reviews related work on textual sentiment analysis and large language models. Section III introduces our EECSA framework, explaining the integration of Emotion-Enriched Contexts. Section IV provides experimental settings, including the evalution metric, datasets, baselines, experimental platform, and hyper-parameters. Section V presents experimental results, comparing EECSA with existing approaches. Finally, Section VI concludes the paper and suggests future research directions.

II. Related Work

A. Textual Sentiment Analysis

Textual sentiment analysis (TSA) [1], [2] is a significant field in linguistics and semantics, gaining increased attention in the era of big data. Significant progress [18]–[20] has been made in textual sentiment analysis using small language models (SLMs) [3]–[5]. Alexander et al. [18] proposed Domain Adaptation through BERT language model (BERT-ADA). Zeng et al. [19] introduced LCF-TSA, a Local Context Focus mechanism addressing traditional methods' limitations. Song et al. [20] proposed an Attentional Encoder Network (AEN) for targeted sentiment classification. Jin et al. [16] integrated dictionary-based explanatory knowledge into LM-based TSA models. Xu et al. [21] proposed Review Reading Comprehension (RRC), transforming customer reviews into a knowledge source. Nazir et al. [22] introduced IAF-LG, an Interactive Attention Fusion network for TSA tasks. Zhao et al. [23] propose a novel approach named Target Word Transferred ABSA that leverages the initial pre-training scheme of SLMs to enhance TSA tasks by transferring target entities into pivot tokens. These SLM-based approaches achieved significant breakthroughs, but the limited parameter size and small-scale pre-training corpora constrain their performance.

Advancements in LLM-based prompting techniques have led to new sentiment analysis solutions based on LLMs [12], [24]–[26]. Scaria et al. [24]introduce InstructABSA, an LLM-based instruction learning paradigm for sentiment analysis that incorporates positive, negative, and neutral examples into task-specific prompts. Zheng et al. [25] proposed a novel instruction tuning method with retrieval-based example ranking. Yang et al. [12] proposed FaiMA, integrating traditional techniques with LLMs for Multi-domain TSA. Ding et al. [26] introduced LLM-CL, a Continual Learning framework addressing domain knowledge variability.

[1]Code of our EECSA framework: https://github.com/albert-jin/EECSA.

In summary, future research in textual sentiment analysis will focus on LLM-based solutions, advanced instruction engineering, and prompt techniques to enhance performance and generalization capabilities while reducing computational costs. Optimizing instruction learning and prompt engineering will be crucial in zero-shot and few-shot settings.

B. Large Language Models

Before the advent of large-scale language models like ChatGPT, smaller pre-trained models such as BERT, RoBERTa, T5 [3]–[5] achieved success in various NLP tasks.

Since ChatGPT [7] was proposed, Large Language Models (LLMs) have revolutionized AI, leveraging vast data and computational power to achieve unprecedented performance. Models such as OpenAI's GPT series, Meta AI's Llama 3 demonstrate significant potential in various realistic applications. LLMs have made breakthroughs in in-context and few-shot learning, using advanced prompt engineering techniques like Chain-of-Thought (CoT) [9], [10], In-context Learning [11], [12], and Reasoning and Acting (ReAct) [13] to enhance performance [15]. Guo et al. [27] developed a data science analysis system based on LLMs. Wang et al. [15] surveyed LLM-based autonomous agents, highlighting their potential in achieving human-level intelligence. Chen et al. [28] explored the role of prompt engineering in optimizing LLM capabilities. Sahoo et al. [29] surveyed prompt engineering approaches. Google [14] reassessed LLM capabilities, noting they cannot simulate human cognition. Kambhampati et al. [30] critiqued views on LLM capabilities, balancing between over-optimism and reductionist perspectives.

Overall, LLMs have achieved significant success by leveraging vast amounts of data and computational power in NLP tasks. In the future, LLMs are expected to continue driving AI development, expanding application areas, and realizing more intelligent information processing systems.

III. EECSA: Emotion-Enriched Contextual Sentiment Analysis

We divided the EECSA framework into two primary stage: 1. *Emotion-Enriched Context Acquisition*, which involves obtaining emotion-enriched constraints, backgrounds, and analogies; 2. *Emotion Context Integration & Sentiment Prediction*, which focuses on integrating the acquired emotion contexts to predict the final sentiment polarity.

A. Emotion-Enriched Context Acquisition

Here, we detail the implementation of the first stage of EECSA, *Emotion-Enriched Context Acquisition*, focusing on obtaining emotion-enriched constraints, backgrounds, and analogies. This includes outlining the inputs and outputs of each module in this stage, as well as the LLM prompting process.

1) Emotion-Enriched Constraints Acquisition: Recent explorations in LLM-based prompting engineering have demonstrated the emergent capability to perform multi-step reasoning for various linguistic tasks, specifically through the Chain-of-Thought (CoT) prompting technique. Although CoT enables more accurate decision-making, the rationality of the reasoning chain is uncontrollable, as the entire reasoning process relies on internally generated ideas, leading to issues like factual hallucinations and error propagation. Similarly, in fine-grained NLP tasks including textual sentiment analysis, LLMs' reasoning and predictions can be influenced by various factors, leading to misjudgments, especially in contexts with less apparent semantic features. Therefore, it is necessary to design linguistic and semantic-level constraints to ensure LLMs systematically complete TSA tasks. Inspired by this, we propose *Emotion-Enriched Constraints*.

Specifically, we designed multiple linguistic variants of the original sentences that require entity sentiment identification, along with a set of corresponding semantic mapping rules. By leveraging linguistic characteristics, we can effectively reduce error propagation and hallucination issues in the sentiment classification procedure of LLMs.

Specifically, as shown in the following text blocks, let's take the example sentence S := "The laptop's design is sleek, ..." and the entities T = "laptop's design" & "battery life" for presentation, we design and obtain LLM responses to the linguistic variants of the queries Q := "`From` S`, can we infer that the` T `is positive/neutral/negative?`". Finally, we use these LLM-generated responses, serving as the obtained emotion-enriched constraints for the subsequent procedure, *Emotion Context Integration & Sentiment Prediction*.

> <PROMPT>
> # Your task is to answer the following questions with "Yes" or "No" and input the answer into the slot ${PRED}.
> # For "laptop's design":
> 1: "From 'The laptop's, ...', can we infer that the laptop's design is positive?" ${PRED}
> 2: "From 'The laptop's, ...', can we infer that the laptop's design is neutral?" ${PRED}
> 3: "From 'The laptop's, ...', can we infer that the laptop's design is negatice?" ${PRED}
> , ...
>
> # Let's solve these questions:
> Copy and paste these problems here, and replace the ${PRED} placeholder with the corresponding label, "Yes" or "No".

This approach ensures that the LLM considers multiple perspectives and reduces errors, enhancing the accuracy and reliability of sentiment predictions in EECSA.

2) Emotion-Enriched Backgrounds Acquisition: In Textual Sentiment Analysis (TSA) tasks, advanced prompting techniques like In-Context Learning (ICL) face several limitations and challenges. For instance, in ICL, the LLM's performance does not heavily depend on the ground truth of input-label

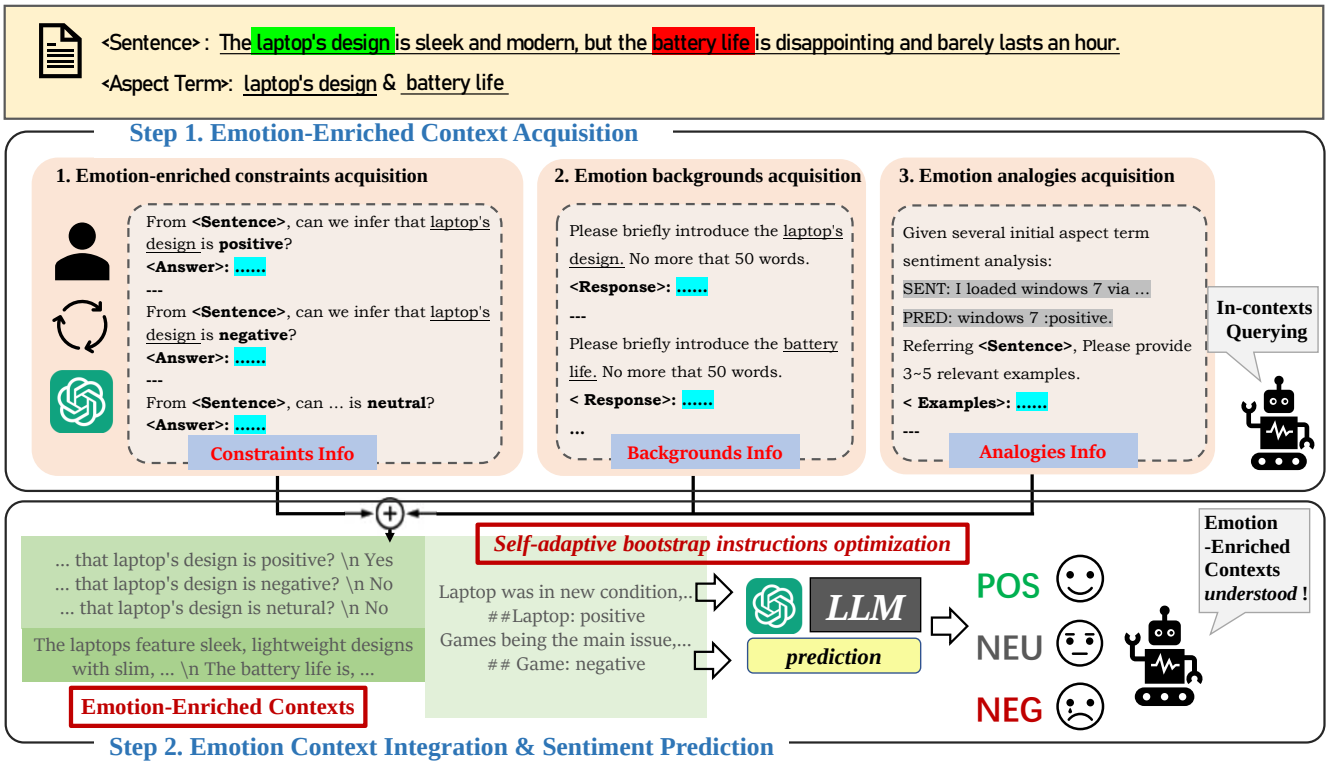


Fig. 1. The overall structure of our proposed EECSA framework. We divided it into two primary stages: **1. Emotion-Enriched Context Acquisition:** We obtain emotional contexts related to the current TSA problem through a series of queries to the LLM, which includes Emotion-Enriched Constraints, backgrounds, and analogies; **2. Emotion Context Integration & Sentiment Prediction:** Based on the contexts obtained from the previous stage, these emotion-enriched contexts are integrated and used for the original TSA task, assisting the LLM in better understanding and predicting sentiment polarity. After integrating the previously generated prompts, we introduced a strategy called *Self-Adaptive Bootstrap Instructions Optimization*, aimed at dynamically adjusting and refining the content and structure of prompts through iterative feedback. This strategy continually optimizes the selection of prompts to enhance the instruction generation and execution efficiency of the language model.

mappings in demonstrations, leading to a marginal drop in performance when labels are randomized. Moreover, ICL's effectiveness is highly influenced by the distribution of the input text, with a significant decline when using out-of-distribution (OOD) text. Therefore, it is crucial to move beyond simple ICL techniques and design more robust, context-aware prompting strategies to ensure accurate TSA by LLMs.

To address these limitations, we introduce the *Emotion-Enriched Backgrounds Integration* strategy for LLM prompting-based TSA tasks. This strategy incorporates external knowledge about the target entities to enhance the efficacy of emotion-enriched contexts in sentiment analysis. This strategy involves prompting LLMs to generate brief introductions for each target entity. These concise feature descriptions, termed emotion-enriched backgrounds, serve as additional prompts to improve sentiment determination.

The process is as follows: First, we design prompts that instruct the LLM to generate background information for the target entities. For instance, given the sentence S := "The laptop's design is sleek, ..." and the entities T = "laptop's design" & "battery life", we craft specific prompts, as illustrated follows.

```
> <PROMPT>
# Given the sentence:
${S} := "The laptop's design is, ..."
, and the target entities ${T} := "laptop's design", and
"battery life".

# Please provide a brief introduction about ${T},
focusing on what makes it positive or negative for
users. No more than 50 words.

# Following the response format for each target entity:
% "${entity 1: intro 1}",
% "${entity 2: intro 2}",
, .....
```

These prompts lead the LLM to generate relevant background information for each entity, as listed follows.

Finally, the generated background information is integrated into the LLM's emotion-enriched contexts, helping the LLM understand the subtleties and specifics of each target entity and leading to more accurate and context-aware sentiment judgments.

> <LLM RESPONSE>
% ”laptop’s design: Laptop design refers to the aesthetics and build quality of the device, often characterized by sleek lines, modern materials, and an overall appealing look that enhances user experience.”
% ”battery life: Battery life in laptops is crucial as it determines how long a laptop can be used without recharging. Poor battery performance can significantly hinder productivity and user satisfaction.”

3) Emotion-Enriched Analogies Acquisition: Inspired by analogical reasoning in psychology, where humans use past experiences to tackle new challenges, we replicate this process in LLMs to enhance their performance in TSA tasks. When faced with a new sentiment analysis task, humans naturally draw upon similar past situations to gain insights. They recognize patterns of positive and negative feedback based on context, which helps in understanding new scenarios.

Here, we prompt LLMs to autonomously generate relevant analogies that provide deeper context for sentiment analysis. Specifically, given a sentence to analyze, we instruct the LLM to recall similar scenarios and describe the sentiments involved, enriching the LLM’s understanding. By integrating these analogies into the LLM’s reasoning process, we significantly improve its ability to perform accurate and contextually aware sentiment analysis.

The process is as follows: We start by designing prompts that guide the LLM to recall and generate relevant analogies. For example, given the sentence S := ”The laptop’s design is sleek, ...” and the entities T = ”laptop’s design” & ”battery life”, we craft specific prompts, as illustrated follows.

> <PROMPT>
Given the sentence ${S} := ”The laptop’s design is sleek, ...” , and the target entities ${T} := ”laptop’s design”, and ”battery life”.
Please recall relevant analogies, and provide similar cases like the above ${S} & ${T}.

Following the response format for each analogy:
”${sentence}”,
”${entity 1: sentiment 1, entity 2: sentiment 2, ...}”,

”${sentence}”,
”${entity 1: sentiment 1, entity 2: sentiment 2, ...}”,
, ...

These prompts are then fed to the LLM, prompting it to generate analogies relevant to the given entities. For example, the LLM might generate responses like:

> <LLM RESPONSE>
% ”A smartphone received high praise for its sleek and innovative design, but many users were disappointed with its short battery life, leading to mixed reviews.”
%% ”smartphone”: positive, ”battery life”: negative.

% ”A car was celebrated for its stylish design and comfortable interior, but it faced criticism due to its poor fuel efficiency.”
%% ”car”: positive, ”fuel”: negative.

Finally, the generated analogies are integrated into the LLM’s emotion-enriched contexts, which involves providing the LLM with these analogies as part of the input context for the original sentiment analysis task.

By following steps, the LLM’s capability is further enhanced to perform nuanced sentiment analysis by leveraging relevant analogies and providing richer contextual information.

B. Emotion Context Integration & Sentiment Prediction

In the second stage, the obtained emotion-enriched contexts are used as the upper context for In-Context Learning (ICL) to assist the LLM in making sentiment predictions. We integrate the Emotion-Enriched Contexts from the first stage with the original sentence S and entities T. These contexts provide a comprehensive background for the sentiment analysis of the original sentence, enabling the LLM to make more accurate and nuanced sentiment judgments.

The process involves the following steps:

1. Processing Emotion-Enriched Constraints using Consistency Alignment Mechanism: To ensure consistent reasoning, if the LLM confirms that ”the laptop’s design is positive/negative/neutral,” then: ”yes/no/no” $\rightarrow$ (positive); ”no/yes/no” $\rightarrow$ (negative); ”no/no/yes” $\rightarrow$ (neutral). Such a consistency alignment mechanism ensures that sentiment judgments are consistent and accurate by verifying the truth values of the constraints and converting them into a unified sentiment prediction. We incorporate these prompts and clues into the context for the final LLM sentiment predictions.

2. Integration of Emotion-Enriched Contexts: The Emotion-Enriched Contexts obtained from the first stage, including processed emotion context constraints from the consistency alignment mechanism, emotion context backgrounds, and emotion context analogies, are integrated as the background for the initial query. For instance, for the sentence S := ”The laptop’s design is sleek and modern, but the battery life is disappointing and barely lasts an hour.” and entities T := ”laptop’s design” and ”battery life”, the gray text blocks here shows an typical example of how such a prompt looks.

By integrating these constraints, backgrounds, and analogies, we provide the LLM with rich and consistent contexts for predicting sentiment. This process ensures that the LLM has a comprehensive understanding of the context, leading to more accurate and reliable sentiment predictions.

> <PROMPT>
${Constraints}:
- From "The laptop's design is sleek and modern, ...", can we infer that the laptop's design is positive? Yes.
- From "The laptop's design is sleek and modern, ...", can we infer that the laptop's design is negative? No.
- From "The laptop's design is sleek and modern, ...", can we infer that the laptop's design is neutral? No.

Note that:
"yes/no/no" → (positive);
"no/yes/no" → (negative);
"no/no/yes" → (neutral).
${Backgrounds}:
- "laptop's design": The design of the laptop is sleek and modern, characterized by its clean lines and contemporary aesthetics.

${Analogies}:
- "laptop's design": A smartphone with a sleek design received positive feedback for its appearance, similar to how the laptop's design is appreciated.

- Given the above contexts, ${Constraints}, ${Backgrounds}, and ${Analogies}, your task is to identify the sentiment polarity of ${T} in ${S}. positive/ negative/ neutral?

C. Self-Adaptive Bootstrap Instructions Optimization

To enhance the performance and robustness of our EECSA framework, we employ a *Self-Adaptive Bootstrap Instructions Optimization* strategy. The primary goal is to systematically fine-tune the prompts and demonstrations provided to LLMs, ensuring they generate more accurate and reliable sentiment predictions. This approach is a bootstrap technique with random search for few-shot optimization, combining the principles of few-shot learning and random search to iteratively refine the instructions and examples provided to the LLM.

As outlined in Algorithm 1, this algorithm operates through the following steps:

First, we initialize essential configuration parameters, including the number of bootstrapped demonstrations, labeled examples, and candidate programs to be generated and evaluated. In each trial, the training set is shuffled to create diverse examples, and a temporary program is initialized. For each stage, a subset of labeled examples is extracted from the shuffled set. Bootstrapped demonstrations are generated by predicting outputs with the current model, validating them against the performance metric, and selecting the top predictions. These demonstrations are combined with labeled examples to form the few-shot examples, updating instructions accordingly. Each candidate program is evaluated and scored. The candidate with the highest score is selected as the optimized program, ready for further tasks or deployment.

Algorithm 1 The pseudo-code procedure of *Self-Adaptive Bootstrap Instructions Optimization* strategy.

Input: initial instruction program P, training set T, validation set V, performance metric M, number of trials N
Output: Optimized program P^*

Initialize configuration parameters: max bootstrapped demos, max labeled demos, num candidate programs, etc.
Initialize an empty list of candidate programs and their scores
for $i = 1$ to N **do**
  Shuffle the training set T
  Initialize a temporary program P_i as a copy of P
  for each stage in P_i **do**
    Extract a subset of labeled examples from the shuffled training set
    Generate bootstrapped demonstrations by:
    - a. Using the current model to predict outputs for the extracted examples
    - b. Validating the predicted outputs against the performance metric M
    - c. Selecting the top predictions that meet the metric criteria
    Combine the bootstrapped demonstrations with labeled examples to form the few-shot examples for this stage
    Update the instructions and examples for the current stage with the newly formed few-shot examples
  end for
  Evaluate P_i on validation set V using metric M
  Record the score of P_i and add it to the list of candidate programs
end for
Select the program P^* with the highest score from the list of candidate programs
return Optimized program P^*

This optimization strategy ensures that the adopted LLM is provided with the most effective and contextually relevant instructions, leading to more accurate and reliable sentiment predictions in the EECSA framework.

IV. Experimental Settings

1) Datasets: For our experimental evaluation, we utilize six well-annotated benchmark datasets, each tagged with target entities and their sentiment polarities, making them ideal for textual sentiment analysis (TSA). An overview of these datasets is provided below:

- **Rest14 & Lap14:** Benchmarks from the SemEval-2014 competition [31], widely referenced in sentiment analysis research for both the restaurant and laptop domains.[2]
- **Rest15:** An extension of the SemEval-2014 restaurant dataset, released by the workshop organizers [32].

[2]SemEval-2014: http://alt.qcri.org/semeval2014/task4/.

- **Rest16:** The SemEval-2016 dataset [33], which continues from previous SemEval-2015 competitions, focusing on the restaurant domain.
- **Twitter:** Compiled and annotated by Mitchell et al., and later reorganized by Li et al. [34]. This dataset includes tweets from the Twitter platform.
- **AclShortData:** This dataset, introduced at ACL 2014 [35], is based on social tweets annotated via Amazon Mechanical Turk.[3]

Table I summarizes the statistical details of these adopted benchmark datasets. For each benchmark, the 'few-shot' column shows the number of training examples used in few-shot learning scenarios (where K = 3, 6, or 9 examples).

TABLE I
THE STATISTICS OF THESE WIDELY-USED TSA BENCHMARKS.

Benchmarks	Train		Dev	Test	Total
	full/initial	few-shot			
Rest14	1796	3/6/9	181	601	2578
Lap14	1308	3/6/9	150	411	1869
Rest15	750	3/6/9	82	400	1232
Rest16	1113	3/6/9	118	420	1651
Twitter	1687	3/6/9	422	233	2342
ACLShortData	5623	3/6/9	625	692	6940

2) Metric: To maintain consistency with previous studies, we use *Accuracy* as the primary evaluation metric as it is a commonly used metric in textual sentiment analysis research. *Accuracy* indicates the proportion of correctly predicted instances out of the total number of instances. It is defined by Eq 1:

$$\text{Accuracy} = \frac{Count_T}{Count_N} \tag{1}$$

Here, $Count_T$ represents the number of correctly predicted instances, while $Count_N$ signifies the total number of instances.

3) Baselines: In our experiments, we evaluate several models as comparative baselines for comparisons with our EECSA, both traditional small language models (SLMs) and recent large language models (LLMs).

The traditional SLMs-based models evaluated in our experiments include: **BERT-SPC** [3], which uses a sentence pair classification task with BERT, constructing input sequences as "[CLS]" + context + "[SEP]" + [target entity] + "[SEP]"; **BERT-BiLSTM-CRF** [36] combines BERT's contextual understanding, BiLSTM's sequence feature extraction, and CRF's sequence labeling; **LCF-ABSA** [19] integrates local and global context features with Semantic-Relative Distance (SRD) and a pretrained BERT model; **BERT-PT** [21] enhances BERT for domain-specific tasks like Review Reading Comprehension (RRC) and TSA using a novel post-training technique; **AEN-GloVe** and **AEN-BERT** [20] use label smoothing regularization with GloVe embeddings and pre-trained BERT to improve performance; **BERT-ADA** [18] employs a two-step procedure: self-supervised domain-specific BERT fine-tuning followed by task-specific fine-tuning.

The adopted advanced LLMs-based models include: **Direct Ask on GPT-3.5-turbo** (*abbr.,* vanilla GPT), using `GPT-3.5-turbo` [6], [37] directly for TSA without additional prompting techniques; **Chain-of-Thought on GPT-3.5-turbo** (*abbr.,* CoT-GPT) employs Chain-of-Thought prompting [9], [10], guiding the LLM through reasoning steps to improve accuracy; and **In-context Learning on GPT-3.5-turbo** (*abbr.,* ICL-GPT) uses In-Context Learning prompting [11], providing example input-label pairs to enhance performance.

4) Hyperparameters: To ensure the reproducibility of our experiments, we used Google Colab, an online cloud platform, to run and share code, offering a convenient environment for conducting experiments and disseminating results. The virtual machine configuration used is as follows: OS: Ubuntu 22.04.3 LTS; CPU: Intel(R) Xeon(R); GPU: Tesla T4; and CUDA Version: 12.2. In our experiments, we set various parameters to optimize EECSA by using the *Self-Adaptive Bootstrap Instructions Optimization*. We configured the maximum number of bootstrapped demonstrations to 4 and the maximum number of labeled demonstrations to 16. The number of training and evaluation rounds was limited to 1. We set the number of candidate programs to 16 and utilized 6 threads for parallel processing. To manage errors, we allowed a maximum of 10 errors. For the selection of few-shot examples, we selected the first K training samples from each category label for K-shot settings (K = 3, 6, and 9). The code implementation of the EECSA framework is available at https://github.com/albert-jin/EECSA.

V. EXPERIMENTAL RESULTS

A. Zero-Shot Results

Table II presents a comparative summary of the proposed EECSA framework's performance against SOTA baselines under zero-shot learning setting. For the implementation of the GPT-based In-Context Learning method (ICL-GPT), we selected three in-context examples from the train set that correspond to the first occurrences of three distinct sentiment categories to serve as the context for this prompting approach.

Models under the SLM-based fine-tuning paradigm are trained on full data, aligning with the same data distribution, which inherently results in superior predictive performance compared to LLM prompting in a zero-shot setting. Therefore, our analysis primarily focuses on comparisons drawn from different methods based on the LLM-based prompting paradigm.

At first glance, the overall results in Table II indicate that our proposed EECSA framework significantly outperforms other baselines, such as CoT-GPT and ICL-GPT, on most benchmarks. This demonstrates the efficacy of our core idea of *Integrating Emotion-Enriched Context* in EECSA for zero-shot sentiment analysis scenarios. Specifically, EECSA excelled on the Rest15 dataset with an accuracy of 82.2%, significantly

[3]ACLShortData: https://github.com/songyouwei/ABSA-PyTorch/tree/master/datasets/acl-14-short-data.

TABLE II

THE PERFORMANCE STATISTICS (METRIC: % ACCURACY) BETWEEN OUR EECSA AND THE SOTA BASELINES UNDER ZERO-SHOT SETTING. THE TOP PERFORMANCE IS HIGHLIGHTED IN BOLD, WHILE SECOND-BEST ARE UNDERLINED. RESULTS WITH ASTERISK (*) ARE SOURCED FROM PRIOR STUDIES.

Methods / benchmarks		Rest14	Lap14	Rest15	Rest16	Twitter	AclShortData
		Acc (%)	Acc (%)	Acc (%)	Acc (%)	Acc (%)	Acc (%)
SLM-based Fine-tuning (trained on full data)							
BERT-SPC		81.0	79.0	88.7	87.3	76.0	73.5
BERT-BiLSTM-CRF		87.3	79.9	**90.3**	88.5	76.4	70.9
BERT-PT*		85.0	78.1	90.0	87.7	-	-
AEN-GloVe*		81.0	73.5	-	-	-	72.8
AEN-BERT*		83.1	79.9	-	-	-	74.7
BERT-ADA*		**87.7**	79.9	**90.3**	88.2	75.8	71.6
LCF-ABSA*		87.1	**82.5**	90.1	88.9	76.6	72.7
LLM-based Prompting (Zero-Shot Reasoning)							
	Samples Count (Test)	*600*	*411*	*400*	*420*	*233*	*691*
Direct Ask on GPT-3.5	Acc (%)	79.20	75.71	79.00	90.24	48.07	49.57
-turbo (vanilla GPT)	Corrected Samples	316	159	316	379	112	343
Chain-of-Thought on	Acc (%)	81.50	77.64	57.00	90.00	44.64	26.16
GPT-3.5-turbo (CoT-GPT)	Corrected Samples	489	315	228	378	104	181
In-context Learning on	Acc (%)	81.50	77.13	78.75	87.62	49.79	50.14
GPT-3.5-turbo (ICL-GPT)	Corrected Samples	489	317	315	368	116	347
Our Method (Zero-Shot Reasoning)							
EECSA (ours)	Acc (%)	**82.14**	**79.02**	**82.20**	**91.50**	**54.90**	**53.54**
(GPT-3.5-turbo)	Corrected Samples	**492**	**324**	**329**	**384**	**128**	**370**

higher than CoT-GPT's 57.00%, marking a 25.2% improvement. On the Rest16 dataset, EECSA achieved an accuracy of 91.50%, surpassing CoT-GPT's 90.00%, indicating a 1.5% improvement. EECSA also performed well on the Twitter dataset with an accuracy of 54.90%, surpassing CoT-GPT's 44.64% by 10.26%. Moreover, EECSA achieved an accuracy of 53.54% on the ACLShortData, which is much higher than CoT-GPT's 26.16%, reflecting a 27.38% improvement.

On the Rest14 and Lap14 datasets, EECSA showed improvements of 0.64% and 1.38% over CoT-GPT, respectively. While these improvements are positive, they are relatively modest. This can be attributed to: Both the Rest14 and Lap14 datasets may contain reviews with straightforward sentiment expressions that CoT-GPT can handle effectively, reducing the added value of Emotion-Enriched Contexts in EECSA. Moreover, the pre-training of `GPT-3.5-turbo` might encompass data similar to the Rest14 and Lap14 datasets. Thus, the distribution patterns and domain-specific terminologies in the Rest14 and Lap14 datasets are better captured by such vanilla GPT-based methods.

On the Rest16 benchmark, we observed that the superior performance of our EECSA, with an accuracy of 91.50%, surpasses both CoT-GPT's 90.00% and Direct Ask on GPT-3.5 (90.24%) as well as ICL-GPT (87.62%). Remarkably, EECSA's accuracy of 91.50% on Rest16 even surpasses several SLM-based fine-tuning methods. For instance, EECSA exceeds BERT-SPC's accuracy of 87.3% by 4.2% and BERT-PT's 87.7% by 3.8%. This highlights that EECSA not only competes well with, but in some cases, outperforms full-data supervised methods that are extensively trained on the same distribution of training data. It demonstrates EECSA's robust performance even when compared to other sophisticated models.

Overall, EECSA exceeds most advanced SLM-based and LLM-based approaches in zero-shot settings across diverse datasets. Its capacity to integrate comprehensive contextual insights and refine prompting instructions positions it as a superior choice for textual sentiment analysis tasks.

B. Few-Shot Results

We assessed the performance variations of our EECSA framework in scenarios with limited labeled data through few-shot learning experiments. By evaluating EECSA against two other baseline models — Vanilla GPT (Direct Ask on GPT-3.5) and CoT-GPT (Chain-of-Thought on GPT-3.5-turbo) — across six benchmarks (Rest14, Lap14, Rest15, Rest16, Twitter, AclShortData) in few-shot settings of $K = \{3, 6, 9\}$, we aimed to determine each method's capability to leverage minimal training data.

Our few-shot learning methodology involved selecting the top K samples for each sentiment category from the sorted datasets. This selection method ensures a balanced sample set, which enhances the model's generalization and ensures fair and reproducible experiments.

Table III provides the performance statistics on accuracy (%) of these LLM prompting-based methods under the different few-shot settings. This table show a consistent phenomenon: EECSA outperforms both Vanilla GPT and CoT-

TABLE III
THE COMPREHENSIVE PERFORMANCE STATISTICS (METRIC: % ACCURACY) BETWEEN OUR EECSA AND THE SOTA BASELINES UNDER FEW-SHOT SETTINGS (K $\in \{3, 6, 9\}$). THE BEST RESULTS ARE IN BOLD, WHILE SECOND-BEST ARE UNDERLINED.

Few-Shot Settings (K-shot)	Models	Rest14	Lap14	Rest15	Rest16	Twitter	AclShortData
K=3	Vanilla GPT	82.83	82.48	83.50	89.52	59.23	62.66
	CoT-GPT	85.33	82.24	79.50	90.24	61.80	46.02
	EECSA (ours)	**87.17**	**85.40**	**83.00**	**92.62**	**63.95**	**70.48**
K=6	Vanilla GPT	85.33	83.45	84.50	89.52	66.95	69.46
	CoT-GPT	84.33	82.48	83.50	90.24	62.23	51.23
	EECSA (ours)	**87.33**	**85.40**	**86.50**	**92.86**	**72.53**	**76.27**
K=9	Vanilla GPT	87.33	84.91	83.75	90.91	68.24	71.35
	CoT-GPT	88.17	84.43	84.50	91.67	72.96	59.62
	EECSA (ours)	**89.50**	**86.37**	**88.25**	**93.10**	**78.54**	**78.29**

GPT across all datasets and K-shot settings. This demonstrates EECSA's superior performance, consistently achieving higher accuracy compared to the other mthods, especially as the number of shots increases.

In the $K = 3$ setting, EECSA shows significant improvements over the other models. For example, on the Twitter dataset, it reaches an accuracy of 63.95%, compared to 59.23% for Vanilla GPT and 61.80% for CoT-GPT. This trend continues in the $K = 6$ setting, where EECSA demonstrates strong performance improvements across all datasets. Notably, it achieves 86.50% on the rest15 dataset, a substantial improvement over the 84.50% and 83.50% recorded by Vanilla GPT and CoT-GPT, respectively. At the highest K-shot setting ($K = 9$), EECSA maintains its superior performance, attaining the highest accuracy of 89.50% on the rest14 dataset and a notable lead on Twitter with 78.54%.

The consistent performance of EECSA can be attributed to several key mechanisms integrated within the framework. Firstly, the use of *emotion-enriched contexts* — including *constraints, backgrounds, and analogies* — provides a richer and more comprehensive context for sentiment analysis. This method effectively mitigates the issues of LLM hallucinations and error propagation, which are common in other methods. Secondly, the self-adaptive bootstrap optimization enhances the model's capability to refine its instructions and examples iteratively, leading to more accurate predictions. This iterative refinement is particularly beneficial in few-shot settings where the training data is limited, allowing the model to adapt and optimize based on the few available examples.

In contrast, Vanilla GPT relies solely on the model's pre-trained knowledge without additional prompting techniques, resulting in lower performance in few-shot settings. It lacks the structured reasoning process provided by CoT-GPT and the rich contextual insights of EECSA. While CoT-GPT improves upon Vanilla GPT by incorporating a chain-of-thought prompting mechanism, it still falls short of leveraging comprehensive contextual insights. The reasoning process, though step-by-step, remains static and can propagate errors from initial steps without correction mechanisms.

In summary, EECSA significantly outperforms both vanilla GPT and CoT-GPT methods on `GPT-3.5-turbo` in few-shot settings across various datasets. These findings underscore the potential of our EECSA framework to address the challenges of sentiment analysis in low-resource settings and pave the way for future relevant research. In future work, we also plan to conduct the *Parameter-Efficient Fine-Tuning* (PEFT) technique on LLMs such as GPT-3.5 to further enhance our EECSA's adaptability to specific datasets.

*C. our EECSA **P.K.** with advanced GPT-4.0*

In this section, we present a comparative analysis of our EECSA framework against the state-of-the-art GPT-4.0 on both the ACLShortData and Twitter datasets, focusing on its application to various complex applications.

The comparative baseline is `Direct Prediction using GPT-4.0` (*abbr.,* vanilla GPT-4), which employs the advanced GPT-4.0 [7] for direct sentiment prediction. In comparison to GPT-3.5 (approximately 175 billion parameters), the state-of-the-art GPT-4.0, with approximately 1.7 trillion parameters—about ten times the size of GPT-3.5—and a significantly larger pre-training corpus, exhibits markedly superior language comprehension and generation capabilities.

Through extensive experiments, we collect the corresponding sentiment polarity prediction distributions and patterns, as depicted in Figure 2 (e.g., a set of heatmaps illustrating the prediction details of these methods). The following insights and conclusions are drawn from this results.

Comparative Performance Analysis on ACLShortData: The ACLShortData benchmark provides a comprehensive benchmark for evaluating sentiment analysis. Our EECSA framework, which integrates Emotion-Enriched Contexts (EEC), demonstrates significant improvements over vanilla GPT-4.0 in this dataset. As shown in the top two heatmaps,

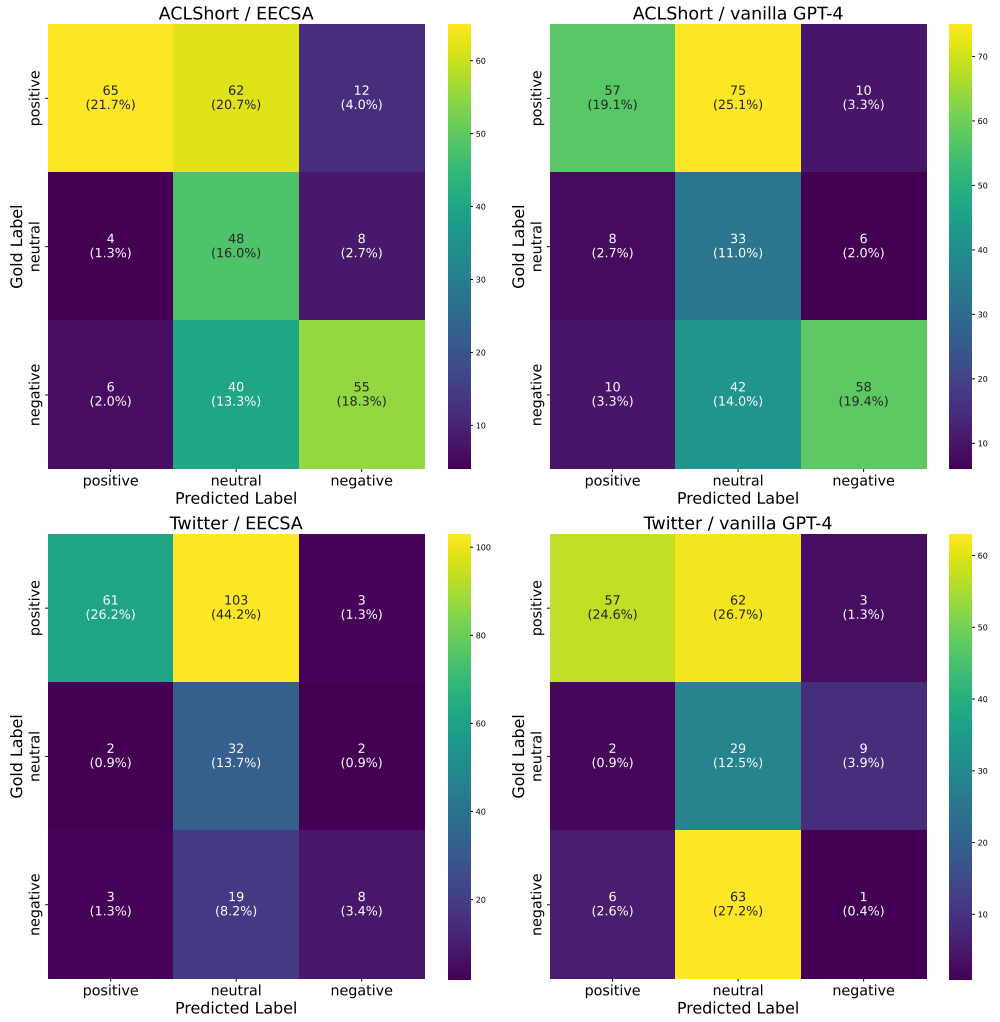


Fig. 2. The heatmaps illustrate the number and percentage of correct and incorrect predictions for each category (positive, neutral, and negative). **Top Left:** prediction distribution using EECSA on AclShortData; **Top Right:** prediction distribution using vanilla GPT-4 on AclShortData; **Bottom Left:** prediction distribution using EECSA on Twitter; **Bottom Right:** prediction distribution using vanilla GPT-4 on Twitter. It can be observed that the EECSA framework generally outperforms the direct use of the GPT-4.0 method across both datasets.

EECSA achieves higher accuracy across all sentiment categories (positive, neutral, negative) compared to GPT-4.0. For positive sentiment, EECSA correctly predicts 65 instances (21.7%) of positive sentiment, outperforming GPT-4.0's 57 instances (19.1%). For neutral sentiment, EECSA also excels in neutral sentiment prediction, with 48 correct instances (16.0%) compared to GPT-4.0's 33 instances (11.0%). This demonstrates EECSA's enhanced capability to understand and interpret positive and neutral emotions. However, for negative sentiment, EECSA predicts 55 instances (18.3%) accurately, while GPT-4.0 manages 58 instances (19.4%). Although GPT-4.0 performs slightly better in this category, EECSA maintains competitive performance with only a small decrease, demonstrating its robustness.

Comparative Performance Analysis on Twitter: The Twitter dataset, known for its concise and informal text, poses unique challenges for sentiment analysis models. As shown in the top bottom heatmaps, despite these challenges, EECSA consistently outperforms GPT-4.0. For positive sentiment, EECSA correctly identifies 61 instances (26.2%) of positive sentiment, surpassing GPT-4.0's 57 instances (24.6%). For neutral sentiment, EECSA achieves an impressive 103 correct predictions (44.2%) for neutral sentiment, significantly higher than GPT-4.0's 62 instances (26.7%). For negative sentiment, EECSA correctly predicts 8 instances (3.4%) of negative sentiment, while GPT-4.0 predicts only 1 instance (0.4%). The integration of emotion-enriched contexts in EECSA significantly enhances its ability to accurately discern sentiments in the often ambiguous context of tweets. This demonstrates EECSA's effectiveness in identifying sentiments, even amidst the noisy and informal language typical of Twitter.

In summary, the above experimental results clearly indicate that our EECSA framework consistently outperforms the vanilla GPT-4.0 in sentiment analysis tasks across both ACLShortData and Twitter datasets. This highlights the importance of integrating emotion-enriched contexts and iterative

optimization strategies in enhancing the performance of LLMs in sentiment analysis. Future research can further explore the potential of combining EECSA with other advanced LLM techniques to achieve even greater performance improvements across various textual sentiment analysis tasks.

VI. Conclusion and Future Work

In this study, we introduce the novel Emotion-Enriched Contextual Sentiment Analysis (EECSA) framework, an innovative approach designed to enhance the TSA task performance using the LLMs-based prompting engineering. Our framework integrates emotion-enriched contexts, including emotion-enriched constraints, backgrounds, and analogies to address the limitations of previous LLM prompting techniques and reduce the impact of LLM hallucinations. By leveraging sophisticated in-context learning (ICL) strategies and self-adaptive bootstrap instructions optimization, EECSA demonstrates superior performance in accurately determining sentiment polarity for target entities within a given sentence. Through extensive experiments with `GPT-3.5-turbo` across six public datasets, we validated the effectiveness and superiority of our framework, even surpassing the reasoning capabilities of GPT-4.0. This underscores the potential of our research to provide more reliable and nuanced sentiment insights, crucial for applications in customer service, marketing, and product development.

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